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Research Interests

Building and urban energy-carbon modeling; HVAC modeling; thermal comfort in large public buildings; HVAC system fault detection and diagnosis; smart building data analytics.

Education

M.Sc., Heating, Gas supply, Ventilation and Air conditioning Works, Sichuan University, Chengdu, China, Sep. 2022-Jun. 2025.

B.S., Building Environment and Energy Application Engineering, Sichuan University, Chengdu, China, Sep. 2018-Jun. 2022.

Selected Publications

- **Jin, Zhineng** and Zhe Wang. "AutoModelling: From BMS Operational Data to Validated Modelica Models for Chillers." *Energy*, 2026, 141700.
- **Jin, Zhineng**, Hongli Sun, Junkang Song, Yanhong Zheng, Hanjie Zheng, Menglong Zhang, and Borong Lin. "Novel integrated meta-sampling approach for smart building design: real-time data analytics and energy performance optimization practice." *Energy*, 2025.
- **Jin, Zhineng**, Hongli Sun, Meng Han, Yanhong Zheng, Ying Zhao, Wenyang Han, Menglong Zhang, Bin Xu, Zequn Zhang, and Borong Lin. "Indoor-outdoor interactive thermal response in public building: onsite data collection and classification through cluster algorithm." *Energy and Buildings*, 2025.
- **Jin, Zhineng**, Yin Zhang, Hongli Sun, Meng Han, Yanhong Zheng, Ying Zhao, Wenyang Han, and Menglong Zhang. "Indoor thermal nonuniformity of atrium-centered public building: Monitoring and diagnosis for energy saving." *Case Studies in Thermal Engineering*, 2024.
- **Jin, Zhineng**, Yanhong Zheng, Dongsheng Huang, Yin Zhang, Siqiang Lv, and Hongli Sun. "Dehumidification load ratio: influence mechanism on air conditioning and energy saving potential analysis for building cooling." *Sustainable Cities and Society*, 2023.
- **Jin, Zhineng**, Yanhong Zheng, and Yin Zhang. "A novel method for building air conditioning energy saving potential pre-estimation based on thermodynamic perfection index for space cooling." *Journal of Asian Architecture and Building Engineering*, 2023.

Conference Presentations

- **Jin, Zhineng**, Si Wu, and Zhe Wang. "Steady-State Detection for Chiller Modelling: A task-grounded benchmark using Modelica-based validation residuals." Oral presentation, Building Simulation BAS 2026, Hong Kong SAR, China, June 11-13, 2026.

Honors and Awards

- Excellent graduate of Sichuan University in 2025.
- National Scholarship for Graduate Students, Ministry of Education, China, 2024.
- Outstanding Graduate Student, Sichuan University, 2024.
- Excellent graduate student of Sichuan University, 2023.
- Comprehensive First-Class Scholarship, Sichuan University, 2021.